Econ 101A
Section 12

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Until now we had one consumer and one firm. Today we build a market. There are three steps. First we add up individual supply and demand curves. Then we find the price that clears the market. Then we ask how that price moves when something changes. Elasticities let us answer the last question without worrying about units, and they also tell us who really pays a tax. Everything today is Lecture 9, first half.

1 Aggregation

Industry supply. With J firms, each with supply $y^j(p, w, r)$, industry supply is

$$Y^S(p, w, r) = \sum_{j=1}^{J} y^j(p, w, r).$$

Market demand. With J consumers, each with income M^j,

$$X^D(p_1, \ldots, p_n, M^1, \ldots, M^J) = \sum_{j=1}^{J} x^j(p_1, \ldots, p_n, M^j).$$

Add quantities, not prices. Fix a price. Ask each agent how much they want at that price. Add the answers. Graphically this is a *horizontal* sum of the curves. Two warnings. Each individual curve can have a kink where the agent starts or stops trading, so the sum has kinks too. And market demand depends on how income is split across people, not only on total income.

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2 Equilibrium in the Short Run

In the short run the number of firms J is fixed. The equilibrium price p^* sets excess demand to zero:

$$Y^S(p^*, w, r) = X^D(p^*, M^1, \ldots, M^J).$$

Nothing stops firms from making positive profits in the short run. That changes in the long run (Section 13).

3 Comparative Statics of Equilibrium

Let a parameter α shift supply, demand, or both. Write equilibrium as an implicit function

$$F(p, \alpha) \equiv Y^S(p, \alpha) - X^D(p, \alpha) = 0.$$

The implicit function theorem gives

$$\frac{dp^*}{d\alpha} = -\frac{F_\alpha}{F_p} = -\frac{\partial Y^S/\partial\alpha - \partial X^D/\partial\alpha}{\partial Y^S/\partial p - \partial X^D/\partial p}.$$

Supply slopes up and demand slopes down, so the denominator is positive. The sign of $dp^*/d\alpha$ is minus the sign of the numerator: anything that raises demand or cuts supply at a given price raises p^*.

4 Elasticities

The size of $dp^*/d\alpha$ depends on units (dollars or cents, tons or pounds). Elasticities remove the units:

$$\varepsilon_{x,p} = \frac{\partial x}{\partial p}\,\frac{p}{x} = \frac{\partial \ln x}{\partial \ln p}.$$

It is the percent change in x for a one percent change in p. Multiplying the comparative statics by α/p and using $X^D = Y^S$ at the equilibrium,

$$\varepsilon_{p,\alpha} = -\frac{\varepsilon_{S,\alpha} - \varepsilon_{D,\alpha}}{\varepsilon_{S,p} - \varepsilon_{D,p}}.$$

This is useful because empirical work gives us elasticities, not slopes.

5 Who Pays a Tax?

A per-unit tax t drives a wedge between the price buyers pay, p, and the price sellers keep, $p - t$. Equilibrium solves $Y^S(p - t) - X^D(p) = 0$. The implicit function theorem gives

$$\frac{\partial p^*}{\partial t} = \frac{\varepsilon_{S,p}}{\varepsilon_{S,p} - \varepsilon_{D,p}}, \qquad \frac{\partial (p^* - t)}{\partial t} = \frac{\varepsilon_{D,p}}{\varepsilon_{S,p} - \varepsilon_{D,p}}.$$

The two numbers are between 0 and 1 in absolute value and add up to one in absolute value. The side of the market that is *less* elastic pays more of the tax. Who hands the money to the government does not matter.

6 Exercises

Exercise 1 (building a market) Each of $J = 10$ identical firms has cost $c(y) = 4 + y^2$ if $y > 0$ and $c(0) = 0$. The 4 is a setup cost the firm avoids by shutting down. There are 5 identical consumers, each with demand $x(p) = \max\{20 - p,\, 0\}$.

(a) Find MC, AC, the minimum of AC, and the supply function of one firm.

(b) Find industry supply $Y^S(p)$. Sketch it. What happens at $p = 4$?

(c) Find market demand $X^D(p)$. A classmate instead adds up the inverse demands $p = 20 - x$ and writes $p = 100 - 5X$. What went wrong?

(d) Find the short-run equilibrium price, total quantity, output per firm, and profit per firm.

Exercise 2 (income distribution matters) Two consumers have Cobb-Douglas utility over good x and a composite good. Consumer j spends a share α_j of income M^j on x, so $x^j = \alpha_j M^j / p$.

(a) Find market demand for x.

(b) Let $\alpha_1 = 0.2$, $\alpha_2 = 0.6$ and total income $M^1 + M^2 = 1000$. Compare market demand when $(M^1, M^2) = (500, 500)$ and when $(M^1, M^2) = (800, 200)$.

(c) Under what condition does market demand depend only on total income?

(d) Supply of x slopes up. Income moves from $(500, 500)$ to $(800, 200)$. What happens to p^*? Use the formula for $dp^*/d\alpha$.

Exercise 3 (comparative statics with the implicit function theorem) Go back to Exercise 1, but let the firm's cost be $c(y) = 4 + w\,y^2$ (so $w = 1$ before) and let demand be $X^D = 100 + \alpha - 5p$ (so $\alpha = 0$ before). Assume the price stays on the upward part of supply.

(a) Find the supply of one firm above its shutdown price. Write the equilibrium condition as $F(p, w, \alpha) = 0$.

(b) Find $dp^*/d\alpha$ with the implicit function theorem. Check it by solving for p^* directly.

(c) Find dp^*/dw and evaluate it at $w = 1$, $\alpha = 0$. Interpret the sign.

(d) At $w = 1$, $\alpha = 0$, compute $\varepsilon_{S,w}$, $\varepsilon_{D,w}$, $\varepsilon_{S,p}$, $\varepsilon_{D,p}$, and use the elasticity formula to get $\varepsilon_{p,w}$. Check with (c).

Exercise 4 (elasticities)

(a) Let $x = k\,p^{\eta}$ with $k > 0$, $\eta < 0$. Find $\varepsilon_{x,p}$ in two ways: with the definition, and by taking logs.

(b) Let $x = a - bp$ with $a, b > 0$. Find $\varepsilon_{x,p}$. At which price is it -1? Evaluate it for $X^D = 100 - 5p$ at $p = 4$, 10 and 16.

(c) With Cobb-Douglas demand $x = \alpha M/p_x$, find the own-price, cross-price and income elasticities.

(d) Empirical work says supply elasticity is 0.5 and demand elasticity is -1. A fad raises demand by 6% at every price. By how much do price and quantity change?

Exercise 5 (who pays the tax?) Use the market of Exercise 1: $Y^S = 5p$ for $p \geq 4$, and $X^D = 100 - 5p$. The government levies a tax of t per unit, collected from sellers.

(a) Find the price buyers pay, the price sellers keep, and the quantity, as functions of t.

(b) Let $t = 4$. Compute all three, the output of each firm, and tax revenue.

(c) Check $\partial p^*/\partial t$ against the elasticity formula at $t = 0$.

(d) Now the tax is collected from buyers instead. Redo (a). What changes?

(e) General linear case: $Y^S = c + dp$, $X^D = a - bp$ with $b, d > 0$. Find $\partial p^*/\partial t$. What happens when $d \to 0$, when $d \to \infty$, and when $b \to 0$?

(f) The payroll tax is split by law: half paid by the employer, half by the worker. Studies find labor supply is quite inelastic and labor demand more elastic. Who really pays?